

Calculus II

Integral of rational function with cubic denominator, part 1

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Example

Find $\int \frac{x^2+2x-1}{2x^3+3x^2-2x} dx$.

- $\deg(x^2 + 2x - 1) < \deg(2x^3 + 3x^2 - 2x)$: don't divide.
- Factor denominator: $2x^3 + 3x^2 - 2x = x(2x - 1)(x + 2)$.

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

$$x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A$$

$$2A + B + 2C = 1$$

$$3A + 2B - C = 2$$

$$-2A = -1$$

Solution:

$$A = \frac{1}{2}, B = \frac{1}{5}, C = -\frac{1}{10}.$$

$$\begin{aligned} & \int \frac{x^2+2x-1}{2x^3+3x^2-2x} dx \\ &= \int \left(\frac{1}{2} \frac{1}{x} + \frac{1}{5} \frac{1}{2x-1} - \frac{1}{10} \frac{1}{x+2} \right) dx \\ &= \frac{1}{2} \ln|x| + \frac{1}{10} \ln|2x-1| \\ & \quad - \frac{1}{10} \ln|x+2| + K \end{aligned}$$

NOTE: There is a quick trick to find A , B , and C .

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

To find A , set $x = 0$; to find B , set $x = \frac{1}{2}$; to find C , set $x = -2$.

$$\begin{aligned} 0^2 + 2 \cdot 0 - 1 &= A(2 \cdot 0 - 1)(0 + 2) \\ -1 &= -2A \\ A &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \left(\frac{1}{2}\right)^2 + 2 \cdot \frac{1}{2} - 1 &= B\left(\frac{1}{2}\right)\left(\frac{1}{2} + 2\right) \\ \frac{1}{4} &= \frac{5}{4}B \\ B &= \frac{1}{5} \end{aligned}$$

$$\begin{aligned} (-2)^2 + 2(-2) - 1 &= C(-2)(2(-2) - 1) \\ -1 &= 10C \\ C &= -\frac{1}{10} \end{aligned}$$

$Q(x)$ has distinct linear factors

- Suppose $Q(x)$ is a product of distinct linear factors:

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_kx + b_k)$$

where no factor is repeated and no factor is a constant multiple of another.

- Then there exist constants $A_1, A_2, \dots, A_k$ such that

$$\frac{R(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \cdots + \frac{A_k}{a_kx + b_k}$$

- We show how to find $A_1, A_2, \dots, A_k$ on examples.